Air Analyzer: A Low-Power Wearable Respiratory Monitoring Device for Athletes

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ABSTRACT

Monitoring respiratory activity during physical exercise is valuable for both performance analysis and health assessment. This work presents the design and implementation of a low-power wearable system integrated into a sports mask, capable of measuring breathing frequency under exertion conditions. The system combines a custom-designed embedded hardware platform, environmental sensing, local data storage, and USB-based data retrieval. The device is fully self-contained, featuring a 3D-printed mask structure and custom firmware. Preliminary experiments highlight the potential of the system for comparing respiratory patterns between trained and untrained individuals.

1 MOTIVATION

Respiratory rate is a key physiological parameter that reflects physical effort, fitness level, and overall health condition. During sports activity, monitoring breathing patterns provides valuable insights into an athlete's performance, fatigue level, and cardiopulmonary adaptation to training.

Research has shown that respiratory rate is a reliable and valid marker for the anaerobic threshold [1], comparable to established ventilatory methods. Unlike heart rate, which requires a stabilization period upon exercise initiation, respiratory rate adjusts almost instantaneously, offering a real-time reflection of the body's exertion level [2]. These properties make respiratory monitoring a particularly informative metric for exercise analysis.

Significant physiological differences exist between trained and untrained individuals. Intense training in elite athletes leads to 10–20% higher forced expiratory volume values compared to the general population [3]. Understanding and quantifying these differences requires portable, unobtrusive monitoring tools capable of capturing respiratory data during real exercise sessions, outside laboratory conditions.

However, existing solutions for respiratory monitoring are often:

- **Intrusive**: chest straps, spirometers, or clinical devices are uncomfortable during dynamic activity;
- **Expensive**: medical-grade equipment is not accessible for everyday sports use;
- **Not suitable for continuous use**: battery life and form factor limit real-exercise deployment.

The goal of this project is to design a wearable, non-invasive, and low-power system integrated into a sports mask, capable of measuring breathing frequency, storing data locally during activity, and enabling post-session analysis of respiratory patterns.

2 LIMITATIONS OF THE STATE OF THE ART

Current respiratory monitoring systems can be broadly classified into two categories.

Medical-grade devices offer high accuracy but are bulky and entirely unsuitable for use during physical exercise. They typically require laboratory settings and trained operators.

Wearable fitness devices (e.g., smartwatches, chest patches) are more portable but often estimate breathing indirectly from heart rate variability or accelerometer data, leading to reduced accuracy, particularly under high motion conditions.

A comprehensive review of wearable respiratory sensors identifies the following recurring limitations [4]:

- **Lack of direct measurement**: many systems do not directly measure airflow or breathing pressure signals;
- **Poor wearability**: devices are uncomfortable during intense activity;
- **High power consumption**: continuous wireless transmission (Wi-Fi, Bluetooth) severely limits battery life;
- **Limited data access**: reliance on proprietary apps or cloud services reduces flexibility for research purposes.

Existing low-power wearable systems for respiratory monitoring have demonstrated that commercial off-the-shelf components are often inefficient for continuous monitoring, both in terms of wearability and energy consumption [?]. Furthermore, most systems are designed for resting or clinical conditions and are not validated for use during intense exercise, where motion artifacts and varying pressure conditions present additional challenges.

A notable recent advancement in mask-integrated respiratory monitoring is the smart facemask proposed by Romano et al. [6], which embeds a temperature biosensor inside a 3D-printed sports mask to measure respiratory frequency (f_R) during cycling exercise. Their system demonstrated excellent accuracy (MAPE of 2.56% and 1.64% breath-by-breath during incremental and HIIT tests, respectively), validating the mask-integration approach for sports applications. However, the system relies on Bluetooth Low Energy (BLE) for continuous wireless data transmission to a Raspberry Pi base station, and requires a lithium coin cell with up to 240 mAh capacity to sustain more than 700 hours of acquisition. Furthermore, it employs a temperature sensor rather than a direct pressure sensor, which means the signal amplitude is strongly influenced by the difference between exhaled air temperature ($T_{Exp} \approx 37\,°C$) and external temperature (T_E): the authors report a substantial reduction in signal amplitude in outdoor conditions above 30 °C (where $T_E \approx T_{Exp}$), which may limit reliability in warm environments.

The system proposed in this work addresses these limitations by combining pressure sensing, local offline data storage that eliminates wireless dependency, and a low-power firmware strategy suitable for coin cell battery operation.

3 KEY INSIGHTS

The key idea of this work is to integrate a barometric pressure sensor directly inside a 3D-printed sports mask, enabling natural and continuous measurement of respiratory activity through detection of the subtle pressure variations caused by inhalation and exhalation.

The main insights that drive the design are:

(1) **Respiratory monitoring does not require real-time transmission.** By storing data locally on EEPROM memory and downloading it after the session via USB-C, it is possible to eliminate continuous wireless radio activity, dramatically reducing power consumption and enabling coin cell battery operation.

(2) **Low-power interrupt-driven firmware reduces active time.** The pressure sensor is configured to acquire data at 10 Hz, while the STM32 avoids continuous polling by using DMA-based SPI communication and interrupt-driven data handling. During idle periods, the main loop executes a wait-for-interrupt instruction, allowing the CPU to remain inactive until a new event must be processed.

(3) **Direct airflow sensing inside the mask.** Placing the pressure sensor inside the mask allows accurate capture of breathing cycles without contact with the skin, improving comfort and signal quality compared to chest-belt or impedance-based approaches.

Together, these design choices provide a balance between usability, accuracy, and system simplicity. Compared to the temperature-based approach of Romano et al. [6], direct barometric pressure sensing offers a signal that is independent of environmental temperature, which is a significant advantage in warm outdoor conditions where the temperature differential between exhaled air and environment is reduced.

4 MAIN ARTIFACTS

The project consists of three main components: hardware, firmware, and mechanical design.

4.1 Hardware Design

The electronic system is built around the following components:

- **STM32L052K8Tx**: ARM Cortex-M0+ ultra-low-power microcontroller, selected for its STOP mode with sub-µA consumption and native USB Full Speed support;
- **LPS22HBTR** (ST Microelectronics): barometric pressure sensor with ±0.1 hPa relative accuracy, 260–1260 hPa range, SPI interface up to 10 MHz, and a built-in 32-level FIFO buffer;
- **M95256**: 256 Kbit (32 KB) SPI EEPROM for local data storage;
- **TPS61291** (Texas Instruments): boost converter with 15 nA quiescent current in bypass mode, stepping up the CR2032 nominal 3 V to a regulated 3.3 V rail;
- **USB-C connector**: for post-session data download to PC.

The schematic sections of the low-power board are illustrated as follow:

- Power supply with boost converter

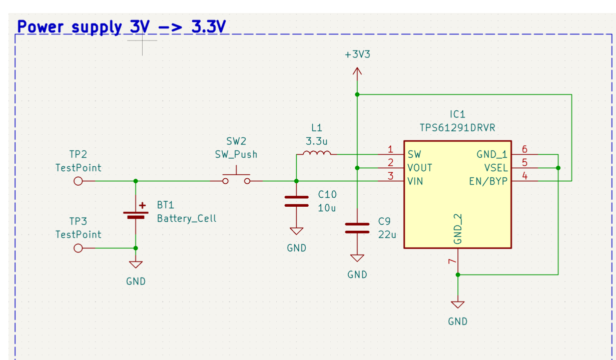

Figure 1: Power supply schematic section necessary to provide 3.3V

- Pressur sensor

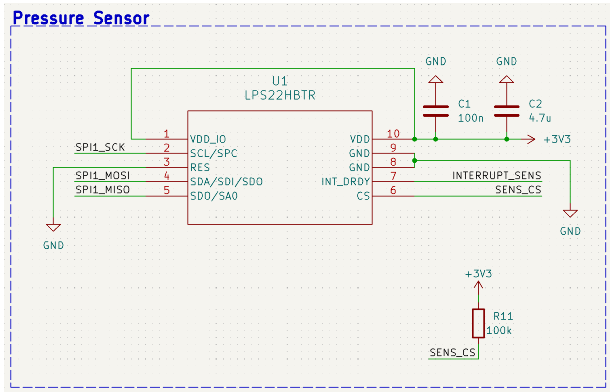

Figure 2: LPS22HB pressur sensor schematic configured

- Microcontroller

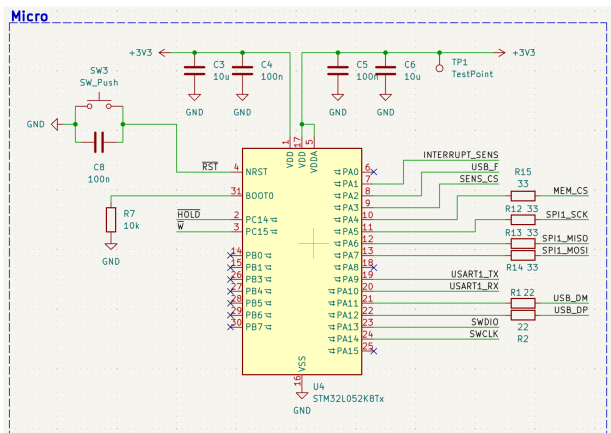

Figure 3: STM32L052 microcontroller with all the surrounding necessary hardware to guarantee to correct operation

- EEPROM memory
- USB-C connector
- UART connector

Cons_ST_STDC14 (ST Microelectronics): 14-pin standard ST debug connector providing:

- SWD interface (SWDIO/SWCLK) for STM32 programming and debugging via ST-Link;
- USART1 TX/RX lines for serial communication with a PC;
- hardware reset line (RST) accessible from the connector;
- VCC pin for powering the board during development, avoiding battery drain.

A custom 50×50 mm two-layer PCB was designed in KiCad 10. Key design decisions include:

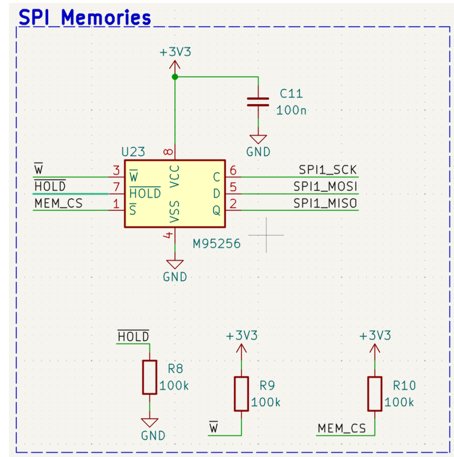

Figure 4: 256Kbit SPI EEPROM for data storage (18 minutes at 10Hz)

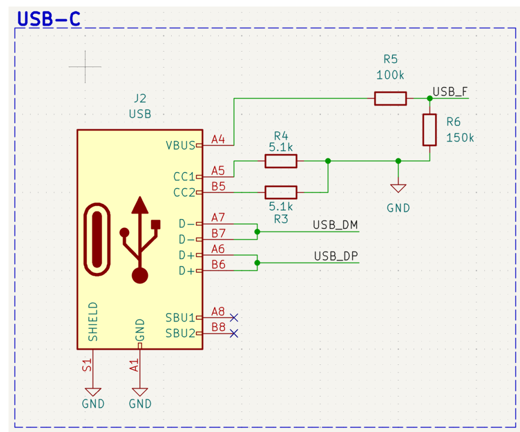

Figure 5: USB-C connector: for post-session data download to PC.

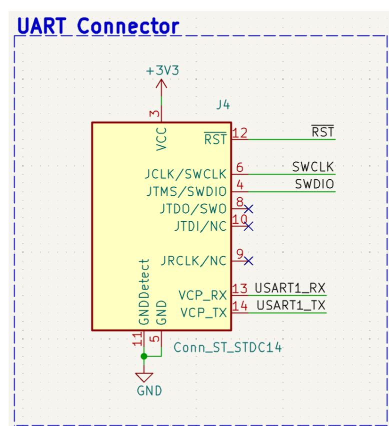

Figure 6: UART connector: for post-session data download to PC.

- 33 Ω series resistors on all SPI lines for signal integrity and EMI damping, placed close to the STM32 output pins;
- decoupling capacitors (100 nF ceramic + bulk) placed at each IC power supply pin;
- ground copper pour on both layers for noise reduction;
- test points on all unused GPIO pins for firmware debugging;
- M2 mounting holes at all four corners for mechanical integration with the mask.

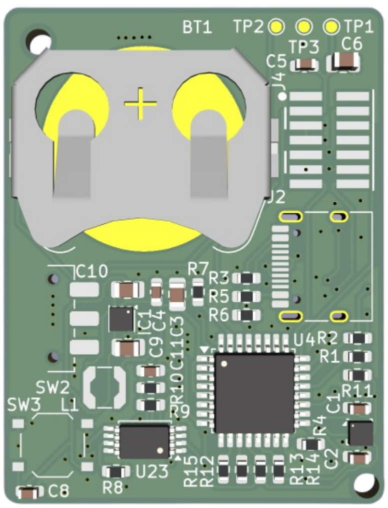

Figure 7: PCB front

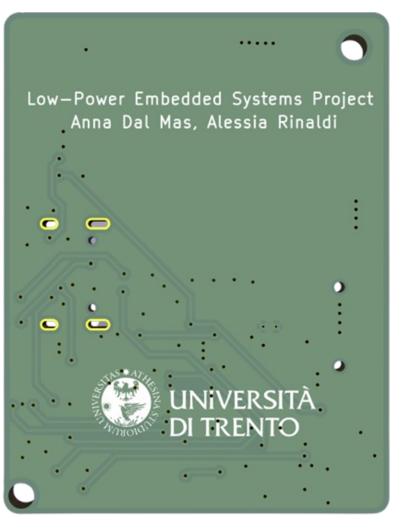

Figure 8: PCB back

4.2 Sampling Strategy and Memory Capacity

Human respiratory rate at rest ranges from 12 to 20 breaths per minute, increasing up to 40–50 breaths per minute during intense exercise [1], corresponding to a maximum signal frequency of approximately 0.8 Hz. Applying the Nyquist–Shannon sampling theorem, the minimum required sampling frequency is:

$$f_s \geq 2 \cdot f_{max} = 2 \times 0.8\,\text{Hz} = 1.6\,\text{Hz} \tag{1}$$

A sampling rate of $f_s = 10\,\text{Hz}$ was chosen to provide ample margin above the Nyquist limit and to capture the full waveform shape of each breathing cycle, enabling analysis of breath depth and respiratory pattern in addition to frequency.

Each pressure measurement occupies 3 bytes (24-bit raw output of the LPS22HBTR. With 32 KB of EEPROM, the maximum recording duration is:

$$T_{max} = \frac{32768\,\text{bytes}}{3\,\text{bytes/sample} \times 10\,\text{Hz}} \approx 1092\,\text{s} \approx 18\,\text{min} \tag{2}$$

4.3 Firmware Implementation

The firmware was developed starting from the STM32CubeMX code generation environment. CubeMX was used to configure the main peripherals of the STM32L052K8Tx microcontroller, including GPIOs, SPI, DMA, USART, USB, and the system clock.

The firmware structure can be divided into three main parts: the initialization code executed once at the beginning of `main()`, the low-power infinite loop, and the interrupt-driven handler used to process the acquired pressure data.

4.3.1 Initialization Code. At startup, the firmware initializes the microcontroller peripherals using the configuration generated with STM32CubeMX. The main configured blocks are the GPIOs, the SPI interface, the DMA controller, and the communication interfaces used for storage and debugging.

The pressure sensor is then configured for continuous acquisition at 10 Hz. Since the respiratory waveform is extracted from pressure variations, the firmware acquires only the pressure signal and stores it in raw format. After this setup phase, the first sensor read operation is started and the acquisition pipeline becomes interrupt-driven.

This initialization stage is executed only once at power-up. Its purpose is to prepare the hardware resources and start the autonomous acquisition mechanism, allowing the main program to remain mostly inactive during normal operation.

4.3.2 Low-Power Main Loop. The infinite loop of the firmware is intentionally kept empty. Instead of continuously polling the sensor or the memory, the microcontroller executes the `__WFI()` instruction, which stands for “Wait For Interrupt”.

This instruction places the CPU in sleep mode until an interrupt occurs.

This design choice is important for reducing power consumption. While the SPI transfer is handled by DMA, the CPU does not need to remain active waiting for the transaction to complete. The processor wakes up only when the DMA transfer-complete interrupt is generated. As a result, the active time of the microcontroller is limited to the short intervals required to manage the acquired sample and prepare the next transfer.

The low-power behavior is therefore obtained through an event-driven firmware architecture: the CPU sleeps during idle periods and resumes execution only when new data must be processed.

4.3.3 Interrupt Handler. The interrupt handler manages the acquisition and storage phase after each DMA-based SPI transfer has been completed. At this point, the pressure sample acquired from the sensor is available in the reception buffer and can be prepared for storage.

The firmware extracts only the useful pressure bytes and stores them directly in raw format. No real-time conversion to physical pressure values is performed on the microcontroller. This choice reduces computational overhead and avoids unnecessary floating-point operations during acquisition. The conversion from raw samples to pressure values is instead performed offline during the post-processing stage.

After each pressure sample is acquired, the firmware prepares a write operation toward the external EEPROM. The stored data correspond to the native 24-bit pressure output of the sensor, so each sample occupies 3 bytes in memory. This compact representation maximizes the available recording duration while preserving the full sensor resolution.

Once the memory operation is completed, a new sensor acquisition is started. In this way, the firmware implements a cyclic acquisition pipeline in which the CPU is active only when a transfer has finished and data must be handled. During the remaining time, the processor can return to a low-power waiting state.

Overall, the interrupt handler is responsible for moving pressure samples from the sensor to the non-volatile memory with minimal CPU intervention. This contributes to the low-power design of the system by combining DMA-based communication, raw data storage, and event-driven execution.

4.4 Mechanical Design

The mechanical structure of the prototype was based on a 3D-printed mask, derived from the open-source *Universal Mask Concept with Seal* model available on Printables [5]. The choice of a 3D-printable mask was motivated by the need for a rigid, reproducible, and easily customizable support capable of hosting the sensing electronics while maintaining a wearable form factor suitable for physical activity.

This structure is particularly suitable for the proposed application because it provides mechanical stability for the electronic system and pressure variations caused by inhalation and exhalation can be detected more clearly by the embedded pressure sensor.

5 KEY RESULTS AND CONTRIBUTIONS

5.1 System Specifications

Table 1 summarizes the key specifications of the Air Analyzer system.

Table 1: Air Analyzer System Specifications

Parameter	Value
Sampling rate	10 Hz
Pressure resolution	1/4096 hPa (≈0.00024 hPa)
Relative accuracy	±0.1 hPa
Measurement range	260–1260 hPa
Recording duration	≈18 minutes
Power supply	CR2032 via TPS61291 boost converter
Operating voltage	3.3 V regulated
PCB dimensions	50×50 mm
Data interface	USB-C (offline, post-session)
SPI clock	up to 10 MHz

5.2 Contributions

This work makes the following contributions:

- **Low-power wearable architecture for respiratory monitoring.** A complete hardware and firmware co-design enabling respiratory pressure monitoring from a CR2032 coin cell battery. Power efficiency is achieved through offline data storage, elimination of wireless radio communication, and FIFO-driven MCU wake-up, minimizing active processor time.
- **Custom PCB and embedded firmware.** A compact 50×50 mm two-layer PCB integrating all system components, designed with full DRC compliance in KiCad 10. The firmware implements a DMA-based, interrupt-driven pipeline with low-power STOP mode, verified to correctly handle SPI communication with both the LPS22HBTR sensor and the M95256 EEPROM.
- **Mask-integrated direct pressure sensing.** Unlike chest-belt or indirect estimation approaches, the sensor is placed directly inside the mask, enabling direct measurement of the pressure variations caused by breathing cycles without requiring skin contact.

- **Platform for athlete vs. non-athlete comparison.** The system provides a simple, low-cost research tool for capturing respiratory data during real exercise sessions, addressing the lack of portable, custom-firmware wearable platforms available for physiological comparison studies between trained and untrained individuals [7].

5.3 Advantages over Past Work

Compared to existing wearable respiratory monitoring systems, Air Analyzer is designed to reduce power consumption and improve experimental flexibility. In contrast to smart-mask approaches relying on Bluetooth or Wi-Fi communication [6], the proposed system stores data locally in external EEPROM, eliminating the continuous power cost associated with wireless transmission and enabling operation from a CR2032 coin cell. Moreover, the use of barometric pressure sensing provides a direct respiratory signal inside the mask, avoiding indirect estimation from heart-rate data and reducing the limitations of thermistor-based approaches, whose performance may degrade in warm outdoor conditions when the environmental temperature approaches the exhaled air temperature. Finally, the open hardware and firmware design makes the platform fully customizable for research purposes, unlike proprietary commercial devices, while the compact 50×50 mm form factor supports integration into a lightweight sports mask.

5.4 Future Work

Future work will focus on the physical fabrication and experimental validation of the board. The first step will involve verifying PCB assembly, power-on behavior, and SPI communication with both the LPS22HBTR pressure sensor and the M95256 EEPROM. Power consumption will then be measured in STOP mode and during active sampling bursts in order to estimate the actual battery life achievable with a CR2032 cell.

The system will subsequently be tested during physical activity to evaluate whether breathing cycles can be reliably detected from pressure variations inside the mask, also considering robustness under motion and sweat conditions. Once the acquisition platform has been validated, respiratory data will be collected from trained athletes and untrained individuals during standardized exercise protocols, following a methodology similar to Romano et al. [6], to assess differences in breathing patterns between the two populations. If the current 22-minute recording window proves insufficient, the M95256 EEPROM may be replaced with a pin-compatible higher-capacity device, such as the M95512 or M95M01, extending the recording time up to approximately 72 minutes without requiring a PCB redesign.

REFERENCES

[1] Daniel G. Carey. Respiratory rate is a valid and reliable marker for the anaerobic threshold. *Journal of Strength and Conditioning Research*, 19(4):899–903, 2005.

[2] CHASKi. Could respiratory rate serve as a superior indicator of physical effort? https://chaski.fit/2024/09/04/could-respiratory-rate-serve-as-a-superior-indicator-of-physical-effort/, 2024. Accessed: 2025.

[3] Marija Filipovic et al. Respiratory parameters in elite athletes – does sport have an influence? *Pulmonology*, 2015.

[4] Tanvir Hussain, Sana Ullah, Raúl Fernández-García, and Ignacio Gil. Wearable sensors for respiration monitoring: A review. *Sensors*, 23(17):7518, 2023.

[5] Petehagoras. Universal mask concept with seal. https://www.printables.com/model/30166-universal-mask-concept-with-seal, 2020. Accessed: 2026-05-18.

[6] Chiara Romano, Andrea Nicolò, Lorenzo Innocenti, Massimo Sacchetti, Emiliano Schena, and Carlo Massaroni. Design and testing of a smart facemask for respiratory monitoring during cycling exercise. *Biosensors*, 13(3):369, 2023.

[7] K. Yamaji and M. Miyashita. Differences in cardio-respiratory responses to exhaustive exercise between athletes and non-athletes. *European Journal of Applied Physiology*, 38(3):233–238, 1978.